IBM BOB: SYSTEM ARCHITECTURE & INTEGRATION BLUEPRINT

High-Availability Cloud Architecture, Agent Orchestration & Box MCP Gateway

Architecture Pattern	Distributed Microservices / Agentic Orchestration Layer / MCP Gateway
Deployment Options	Multi-Tenant SaaS (Isolated VPCs), Single-Tenant Dedicated, Sovereign GovCloud
High Availability Target	99.95% SLA with Multi-AZ Active-Active Failover

1. Architectural Topology Overview

The IBM Bob platform comprises four decoupled, highly scalable architectural layers:

- **Client & Presentation Tier:** Web UI, IDE extensions, CLI tools, and embedded Salesforce/Box UI components communicating over TLS 1.3.
- **Agent Orchestration Engine:** Multi-agent coordinator managing task decomposition, reasoning loops, evidence synthesis, and subagent delegation.
- **Model Context Protocol (MCP) Gateway:** Secure bridge managing bidirectional communication between Bob agents and enterprise data sources like Box Content Cloud.
- **watsonx Enterprise Foundation Layer:** Enterprise model routing supporting IBM Granite, specialized fine-tuned financial models, and customer-hosted LLMs with zero data retention.

2. Data Flow & Security Boundary

When a user requests an RFP response, the agent queries Box via MCP. Box returns encrypted document chunks. Bob performs in-memory reasoning, validates evidence citations against the document hash, and renders the output deliverable. No document content is written to disk or used in background retraining.